

3.5.4 Binary number system

3.5.4.1 Unsigned binary

Content	Additional information
Know the difference between unsigned binary and signed binary.	Students are expected to be able to convert between unsigned binary and decimal and vice versa.
Know that in unsigned binary the minimum and maximum values for a given number of bits, n , are 0 and $2^n - 1$ respectively.	

3.5.4.2 Unsigned binary arithmetic

Content	Additional information
Be able to: • add two unsigned binary integers • multiply two unsigned binary integers.	

3.5.4.3 Signed binary using two's complement

Content	Additional information
Know that signed binary can be used to represent negative integers and that one possible coding scheme is two's complement.	This is the only representation of negative integers that will be examined. Students are expected to be able to convert between signed binary and decimal and vice versa.
Know how to: • represent negative and positive integers in two's complement • perform subtraction using two's complement • calculate the range of a given number of bits, n.	

3.5.4.4 Numbers with a fractional part

Content	Additional information
Know how numbers with a fractional part can be represented in: • fixed point form in binary in a given number of bits.	
Be able to convert for each representation form: • decimal to binary of a given number of bits • binary to decimal of a given number of bits.	